

# Learning Objectives

## 1. Transmission Media

By the end of this topic, students will be able to:

- Define **transmission media** and explain its role in computer communication.
- Classify transmission media into **wired (guided)** and **wireless (unguided)** media.
- Describe the working and characteristics of **twisted-pair cable, coaxial cable, and fiber-optic cable**.
- Compare different wired media based on **speed, bandwidth, cost, range, reliability, and susceptibility to interference**.
- Explain the basic characteristics and applications of **radio waves, microwaves, and infrared waves**.
- Differentiate between wired and wireless transmission media.
- Select an appropriate transmission medium for a given real-world networking situation.

## 2. Network Devices

Students will be able to:

- Identify commonly used networking devices and components.
- Explain the purpose and basic functioning of:
  - **Modem**
  - **Ethernet Card**
  - **RJ45**
  - **Repeater**
  - **Hub**
  - **Switch**
  - **Router**
  - **Gateway**
  - **Wi-Fi Card**
- Differentiate between **hub, switch, router, repeater, and gateway** based on their functions.
- Explain how networking devices facilitate communication between computers and networks.
- Identify suitable networking devices for specific networking requirements.
- Analyse a simple network scenario and determine which device(s) should be used.

## 3. Network Types

Students will be able to:

- Define a **computer network** and explain the need for networking.
- Classify networks according to geographical coverage.
- Explain the characteristics and applications each type
- Compare PAN, LAN, MAN, and WAN in terms of **coverage area, ownership, purpose, and applications**.

- Identify the appropriate type of network for a given real-life situation.
- Analyse scenarios and classify them as PAN, LAN, MAN, or WAN.

## 4. Network Topologies

Students will be able to:

- Define **network topology**.
- Explain the concept of physical arrangement of computers and networking devices.
- Describe and identify:
  - **Bus topology**
  - **Star topology**
  - **Tree topology**
- Draw and interpret basic diagrams of different network topologies.
- Compare network topologies based on **structure, advantages, limitations, scalability, and failure points**.
- Select an appropriate topology for a given networking requirement.
- Analyse the effect of failure of a cable or networking device in different topologies.
- Differentiate between **Bus, Star, and Tree topology** using real-world examples.

## Higher-Order Learning Outcomes

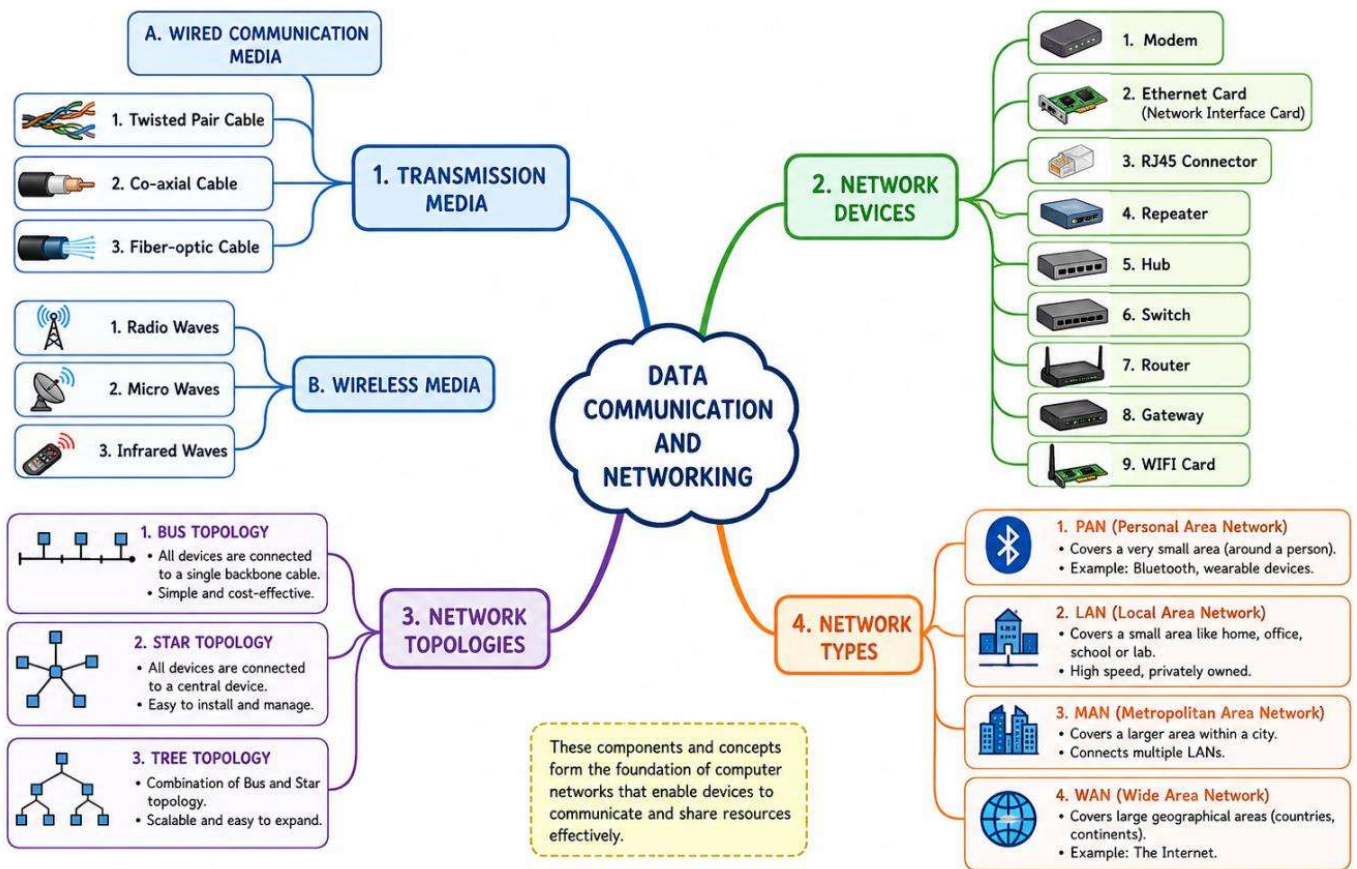
After completing the unit, students should be able to:

- **Analyse** a given network diagram and identify its media, devices, topology, and network type.
- **Compare and evaluate** alternative networking solutions for a given situation.
- **Apply** networking concepts to real-world environments such as schools, offices, homes, and metropolitan networks.
- **Justify** the selection of a particular transmission medium, networking device, network type, or topology.
- **Interpret network diagrams** and identify possible points of failure.
- **Solve competency-based problems** involving network design and device selection.

## Learning Outcome Summary

| Topic              | Students will be able to...                                                       |
|--------------------|-----------------------------------------------------------------------------------|
| Transmission Media | Classify, describe, compare, and select suitable wired/wireless media             |
| Network Devices    | Identify devices, explain their functions, and select appropriate devices         |
| Network Types      | Classify networks as PAN, LAN, MAN, or WAN                                        |
| Network Topologies | Identify, draw, compare, and evaluate Bus, Star, and Tree topologies              |
| Overall Networking | Analyse real-world network scenarios and justify appropriate networking solutions |

## MIND MAP



★ Note: Diagrams are indicative.